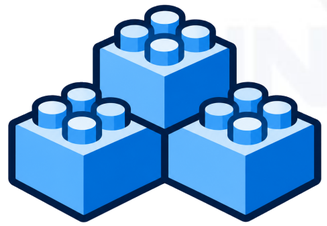
MLOps Specialization Bootcamp

Capstone

End-to-End MLOps

The project brief: what you build by Week 14, and how it is graded



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By Week 14 you will



1. Join every weekly part into one system that runs from a fresh clone
2. Deploy it to AWS with CI/CD, and watch it with monitoring
3. Show it stopping a bad change, and recovering from a bad model
4. Explain and defend your design in a live demo
5. Own a portfolio project you can show an employer

The brief

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01

What you build

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The capstone in short

What it is

- A machine learning system another person can clone, run and trust
- It serves predictions in the cloud, watches itself, and fails safely

On the diamond project

- Data checks, a tracked `champion`, a scheduled pipeline, an API on AWS Lambda
- A dashboard, a drift report, a retrain decision, and a rollback

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Required parts, 1 of 2

Part	Tools	Week
Validated and versioned data	pandera, DVC	3
Tracked experiments, a registered `champion`	MLflow	4
An orchestrated training pipeline	Prefect	5
A tested API and a batch job	FastAPI, Pydantic	6
A container	Docker	7
CI with a quality gate, and CD	GitHub Actions	9

Required parts, 2 of 2

Part	Tools	Week
Infrastructure in Terraform on AWS	Terraform, Lambda, S3	10
Monitoring: a drift report and a dashboard	Evidently, Prometheus, Grafana	11
A feedback loop: champion vs challenger	MLflow, Prefect	12
A rollback	MLflow aliases, CD	12
`docs/architecture.md`: a diagram and a reason per tool	Markdown, Mermaid	Your own

You already build most of it

- Each weekly practical adds one required part to your diamond project
- Finish the 12 practicals, and parts 1 to 10 are ready to connect
- Week 13 is for connecting, testing and your own feature, not for starting
- Fall behind? Check out the next `start-mXX` branch and keep going

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Two tracks

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Track A or Track B

Track A (default)

- The diamond system, extended
- Uses your weekly work
- Add one feature of your own
- Recommended for most students

Track B (approval needed)

- A dataset of your choice
- Same rubric, plus one own feature
- Show the data is legal to use
- Model trains in under 5 minutes

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Your own feature: ideas

- **Shadow deployment:** a second model gets the same requests; its answers are only logged
- **Feature store:** serve the diamond features with Feast (Week 3 demo)
- **Kubernetes:** deploy with a rolling update (Week 8)
- **Front end:** a small Streamlit page that calls your API
- **Cost dashboard:** what your AWS resources cost each week
- **Your idea:** propose it in Week 5

03

Timeline

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Six dates to remember

When	You do	You get
Week 1	Read the brief and the rubric	This brief
End of Week 5	Submit a 1-page proposal	
Week 6	Fix the scope if asked	Scope approved
Week 10	15-minute check-in, live or recorded	Feedback on risks
Week 13	Build week. Bring questions	Office hours, full answer-key demo
Week 14	Freeze, fresh-clone run, demo day	Your grade

The proposal: one page, four headings

1. **Track:** A or B. For B: dataset, link, license, size, training time
2. **Own feature:** what it does, why it helps, how you will demo it
3. **Architecture sketch:** data, pipeline, registry, API, cloud, monitoring
4. **Risks:** the 3 most likely problems, and your plan for each

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Demo day

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Demo day: 10 minutes each

- **Before the session:** a reviewer runs your tagged repo from a fresh clone
- **6 minutes, demo:** the system end to end
  - Then one failed quality check, and a rollback to the previous `champion`
- **4 minutes, questions:** trade-offs in your design
  - Plus one small change on request, which you run live

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A failed check, then a recovery

Failed quality check (examples)

- Validation stops bad data before training
- CI blocks a pull request with a worse model
- The gate keeps a weaker challenger out

Recovery action (examples)

- Roll back to the previous `champion`
- Redeploy the previous image
- Retrain on fixed data, then promote

05

How it is graded

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The rubric: 100 points

Building it

- Reproducibility: 15
- Data quality and traceability: 15
- Automation and delivery: 15
- Monitoring and recovery: 20

Showing it

- Own feature: 10
- Documentation: 10
- Explanation and walkthrough: 15

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06

Rules

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Four rules

- **AI tools are allowed.** You own every line. You must explain it and change it live
- **AWS cost:** your own account, budget alert on, `terraform destroy` after each session
- **No secrets in Git:** no access keys, no `.env` files
- **Your own work:** share ideas; name adapted code in `docs/architecture.md`

This week

- Read `capstone/brief.md` and `capstone/rubric.md`
- Create `docs/architecture.md` from your Practical 1 lifecycle diagram
- Write down one own-feature idea
- Questions: bring them to Week 2, or ask in the course channel

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THANK YOU



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